ROBIN MALIK

Senior Full Stack Software Engineer | Python · Django/DRF · FastAPI · Angular/React

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PROFESSIONAL SUMMARY

Senior Full Stack Engineer with 4+ years building high-throughput, data-intensive platforms in banking/fintech and SaaS. Own a Django/DRF reconciliation engine for bank and card-scheme (NPCI RuPay) operations processing 10M+ transactions/day, from schema design to release, and lead a 4-engineer backend team as feature owner. Comfortable across the stack, from ETL pipelines, Celery/Redis async processing, PostgreSQL tuning and AWS to Angular and React front ends, and currently building an AI-powered reconciliation platform with RAG and agentic tool-calling.

TECHNICAL SKILLS

Languages: Python, SQL, TypeScript, JavaScript
Backend: Django, Django REST Framework, FastAPI, Pydantic v2, Celery, REST API design, Microservices, OpenAPI/Swagger
Data & Messaging: PostgreSQL, Redis, Kafka, ETL pipelines, pgvector
Frontend: Angular, RxJS, React.js, React Query
Cloud (AWS): EC2, S3, RDS, Lambda, API Gateway
Security & DevOps: JWT, RBAC, SSO (Azure AD), Docker, Docker Compose, Nginx, Git, GitHub Actions, Linux
Engineering & AI: OOP, SOLID, design patterns (Adapter, Factory, Strategy), RAG, embeddings, agentic tool-calling (ReAct)

PROFESSIONAL EXPERIENCE

Senior Full Stack Software Engineer, Bancapp Automation **Aug 2024 – Present**

Reconsyde: reconciliation and exception-management platform for banking and card-scheme operations

- Own a Django/DRF data platform and reconciliation engine processing 10M+ transactions/day, targeting a 97.5% auto-match rate.
- Feature owner for reconciliation engine and audit-trail modules: plan work, review code, lead releases for a 4-engineer team.
- Built an ETL pipeline ingesting, transforming and loading feeds from 8 upstream sources into the reconciliation engine; cut ingestion failures with Pydantic-driven schema validation.
- Architected Celery + Redis async infrastructure for reconciliation jobs, scheduled tasks and batch processing: 24 concurrent workers with exponential backoff and idempotent retries.
- Optimized DRF APIs and PostgreSQL query plans (indexing, Redis caching, N+1 elimination), cutting API latency by 85% at peak load.
- Built failure-safe rollback, retry and dead-letter handling to keep high-volume financial workflows atomic and recoverable.
- Built Angular dashboards for real-time visibility into match rates, exceptions and backlog, cutting exception resolution time by 60%.
- Shipped an Exception Management System with audit tracking and remark-driven resolution flows, used daily by 35 operations users.
- Designed JWT authentication with module- and action-level RBAC for multi-role access across all protected platform APIs.
- Automated NPCI RuPay card-scheme email workflows (SPOC notifications for incentive payout, co-brand invoice, cashback, fees and lounge utilization); built the license-management, escalation-matrix and image-analysis modules.

Full Stack Engineer, Alcor Solutions **May 2022 – Jul 2024**

High5 multi-tenant SaaS platform; Volkswagen (VFJ) enterprise client project

- Built and optimized Django REST Framework APIs, validation layers and business logic for High5, a multi-tenant SaaS employee-recognition platform serving 1,200+ enterprise customers, with tenant-aware data isolation.
- Integrated third-party voucher APIs to automate issuance and redemption, cutting processing time ~35% (12s to 7.8s).
- Implemented Azure Active Directory SSO for enterprise onboarding, bringing tenant self-onboarding under one day.
- Improved dashboard responsiveness by introducing React Query caching patterns, reducing repeat API calls.
- Delivered a production Angular client portal for Volkswagen (VFJ) with reusable components, validation and role-aware routing.

PROJECT

AI-Powered Reconciliation Platform (personal project) **2026**

Python, Django/DRF, FastAPI, PostgreSQL, pgvector, Kafka, Celery, Docker

- Building the same domain as a Django (DRF) monolith and as FastAPI microservices to compare architectural trade-offs.
- Designed a multi-source ingestion layer using Adapter and Factory patterns; validated a fixed-width bank GL adapter end-to-end against a 20,000+ row production-style ledger export, normalized into a Pydantic-validated canonical contract.
- Designing an event-driven Kafka pipeline and a Strategy-pattern match engine (exact, fuzzy, embedding-based).
- Applying RAG (chunking, embeddings, pgvector retrieval) and a hand-rolled ReAct tool-calling agent to investigate unmatched transactions with human-in-the-loop review; containerized the full stack with Docker Compose.

TrueKey: Verified Flat-Rental Platform (side project, in development) **2026**

- Building a rental platform with verified listings, owners and tenants, after moving across 10 flats in 8 years.
- Designing couple-friendly search options, locality safety information, in-app rent payments and easy check-ins.

EDUCATION & CERTIFICATIONS

B.Tech, Dr. A.P.J. Abdul Kalam Technical University **2022**

Certifications: Python 3 Programming Specialization (Coursera) · Angular: The Complete Guide (Udemy) · Web Development (Udemy)